BINCY MANUEL

AI/ML Engineer | Machine Learning • Deep Learning • LLMs • RAG • Python

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PROFESSIONAL SUMMARY

AI/ML Engineer with an Integrated M.Sc. in Computer Science (Data Science) and hands-on experience building machine learning and deep learning solutions, including LLM-powered applications, Retrieval-Augmented Generation (RAG) pipelines, AI agents and intelligent automation workflows. Proficient in Python, TensorFlow, PyTorch, Scikit-learn and LangChain, with experience in vector databases, prompt engineering, model evaluation and Docker-based deployment across NLP, computer vision and anomaly detection.

TECHNICAL SKILLS

Programming: Python, SQL, R, Java, C, C++, Kotlin
Machine Learning: Scikit-learn, Feature Engineering, Model Evaluation, Anomaly Detection, ML Pipelines
Deep Learning: TensorFlow, PyTorch, CNN, RNN, Transformers, Natural Language Processing (NLP), BioBERT, Computer Vision
Generative AI: LLMs, RAG, LangChain, Prompt Engineering, AI Agents, Vector Databases, Embeddings
Data & Databases: Pandas, NumPy, EDA, Data Preprocessing, MySQL, PostgreSQL, MongoDB, SQL Server, Power BI, Tableau
MLOps & Tools: Docker, Git, GitHub, Linux, Jupyter Notebook, VS Code, Microsoft Azure, Matplotlib

PROFESSIONAL EXPERIENCE

Data Analyst *Sep 2026 – Present*
Institute of Small Enterprises and Development (ISED) — Vennala, Kerala

- Managing the data pipeline and structuring the workflow for the practical implementation of the 2016 ‘Entrepreneurship Development in Kerala’ report.
- Conducting initial data exploration and requirement gathering to prepare datasets for standalone analysis.

Research Intern *Dec 2025 – Mar 2026*
Indian Institute of Science Education and Research (IISER) — Hybrid

- Conducted 6G network security research on anomaly detection, attack mitigation and intelligent traffic analysis using machine learning and statistical modeling.
- Implemented entropy-based traffic analysis to detect abnormal network behavior and potential cyber threats in real time.
- Developed machine learning and statistical threat-detection mechanisms to improve network resilience and security response.
- Analyzed 500 network traffic datasets with Python, cutting report turnaround time by 20%.

Data Science Intern *May 2024 – Jun 2024*
Null Class — Remote

- Collected, cleaned and pre-processed structured datasets using Python and SQL for machine learning applications.
- Performed exploratory data analysis (EDA) and feature engineering to improve data quality and model efficiency.
- Built analytical dashboards and automated reporting pipelines, optimizing SQL queries to cut query execution time by 30%.

PROJECTS

Data Engineering Agent | *LLMs, AI Agents, Text-to-SQL* — Building an AI agent that converts natural language to SQL and auto-diagnoses and fixes failing data pipelines, improving query accuracy via a self-correction loop.

RAG-Based Telecom Customer Support Chatbot | *LLMs, LangChain, Embeddings, Vector DB* — Developed a RAG chatbot to automate customer support and FAQ resolution, grounding responses to reduce hallucination.

Animal Disease Detection Using BioBERT | *Deep Learning, NLP, Transformers* — Developed a BioBERT-based disease prediction model for healthcare text classification (published, ICAIBT 2025).

Adaptive Edge Security for 6G | *Machine Learning, Anomaly Detection* — Designed AI-driven monitoring workflows for anomaly detection and adaptive threat analysis in 6G networks.

Road Traffic Monitoring System | *Computer Vision* — Real-time vehicle and pedestrian analysis using image processing.

Time Table Generator | *Genetic Algorithms, CSP* — AI-based scheduling and resource-allocation system (published, 2025).

EDUCATION

Integrated M.Sc. in Computer Science (Data Science) *Sep 2021 – Apr 2026 (Completed)*
Sacred Heart College, Thevara, Kerala, India | CGPA: 4.57 / 5.00

Higher Secondary Education *Jun 2019 – May 2021* | Mary Matha H.S.S, Thrikkakara, Kerala | Score: 98.75%

ACHIEVEMENTS

- Qualified UGC NET (June 2026) in Computer Science and Applications — eligible for Ph.D. admission.
- Awarded the India AI Fellowship.

PUBLICATIONS

- Animal Disease Detection Using BioBERT: A Deep Learning Approach — TECHSPARK, ICAIBT, 2025
- Timetable Generator Using Genetic Algorithm and CSP — International Journal for Multidisciplinary Research, 2025

CERTIFICATIONS

Machine Learning Pipelines (Microsoft Azure) • Generative AI • Cyber Forensics • UI/UX Designing